

1) Ex 4-2-1 $J(x) = \int_0^{\pi/2} [\dot{x}^2(t) - x^2(t)] dt$, B.C: $x(0)=0$ & $x(\pi/2)=1$

Euler eq:
$$0 = \frac{\partial g}{\partial x}(x^*(t), \dot{x}^*(t), t) - \frac{d}{dt} \left[\frac{\partial g}{\partial \dot{x}}(x^*(t), \dot{x}^*(t), t) \right]$$
$$= \frac{\partial}{\partial x} [\dot{x}^2 - x^2] - \frac{d}{dt} \left[\frac{\partial}{\partial \dot{x}} (\dot{x}^2 - x^2) \right]$$
$$= -2x^*(t) - \frac{d}{dt} [2\dot{x}^*(t)]$$
$$= -2\dot{x}^*(t) - 2\ddot{x}^*(t) = 0$$
$$\Rightarrow \ddot{x}^*(t) + x^*(t) = 0$$

Let $x^*(t) = k e^{qt}$

$k q^2 e^{qt} + k e^{qt} = 0 \Rightarrow q^2 + 1 = 0 \therefore q = \pm i$

$x^*(t) = c_1 e^{-it} + c_2 e^{+it}$

$\Rightarrow x^*(t) = c_3 \cos t + c_4 \sin t$

B.C: $x(0)=0$ $x(\pi/2)=1$ $\therefore 0 = c_3 \cos(0) + c_4 \sin(0) \Rightarrow c_3 = 0$

$1 = c_4 \sin(\frac{\pi}{2}) \rightarrow c_4 = 1$

$\therefore x^*(t) = \sin(t)$

Ex 4-2-2

smallest length connecting the point $x(0)=1$ to the line $t=5$
 $t_0=0$, $t_f=5$

$$J(x) = \int_0^5 [1 + \dot{x}^2(t)]^{1/2} dt$$

Euler equation

$$\begin{aligned} 0 &= \frac{\partial g}{\partial x} (x^*(t), \dot{x}^*(t), t) - \frac{d}{dt} \left[\frac{\partial g}{\partial \dot{x}} (x^*(t), \dot{x}^*(t), t) \right] \\ &= \frac{\partial}{\partial x} (1 + \dot{x}^{*2})^{1/2} - \frac{d}{dt} \left(\frac{\partial}{\partial \dot{x}} (1 + \dot{x}^{*2})^{1/2} \right) \\ &= 0 - \frac{d}{dt} (\dot{x}^* (1 + \dot{x}^{*2})^{1/2}) \\ &= - \left\{ \ddot{x}^* (1 + \dot{x}^{*2})^{-1/2} + \dot{x}^* \left(-\frac{1}{2} \right) (2 \dot{x}^* \ddot{x}^*) (1 + \dot{x}^{*2})^{-3/2} \right\} \\ &= 0 \end{aligned}$$

$$-\ddot{x}^* (1 + \dot{x}^{*2}) - \dot{x}^{*2} \ddot{x}^* = 0$$

$$\Rightarrow \ddot{x}^* = 0$$

$$\therefore x^*(t) = c_1 t + c_2$$

$$x^*(0) = 1 \rightarrow c_2 = 1$$

$x(t_f)$ is free, $\delta x(t_f)$ is arbitrary

$$\frac{\partial g}{\partial \dot{x}} (x^*(t_f), \dot{x}^*(t_f), t_f) = 0$$

$$\frac{\dot{x}^*(5)}{[1 + \dot{x}^{*2}(5)]^{1/2}} = 0 \rightarrow \dot{x}^*(5) = 0$$

$$\dot{x}(t) = c_1 = 0$$

$$\therefore x^*(t) = 1$$

3) Ex 4.24

$$J(x) = \int_1^{t_f} [2x(t) + \frac{1}{2} \dot{x}^2(t)] dt$$

$$B.C: x(1) = 4, x(t_f) = 4 \text{ \& } t_f > 1$$

Euler equation

$$\frac{\partial g}{\partial x}(x^*(t), \dot{x}^*(t), t) - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}}(x^*(t), \dot{x}^*(t), t) \right) = 0$$

$$(2) - \frac{d}{dt}(\dot{x}^*) = 0$$

$$\rightarrow \ddot{x}^* = 2$$

$$\therefore \ddot{x}(t) = t^2 + C_1 t + C_2$$

t_f is unspecified

$$0 = g(x^*(t_f), \dot{x}^*(t_f), t_f) - \left[\frac{\partial g}{\partial \dot{x}}(x^*(t_f), \dot{x}^*(t_f), t_f) \right] \dot{x}^*(t_f)$$

$$= 2x^*(t_f) + \frac{1}{2} \dot{x}^{*2}(t_f) - (\dot{x}^*(t_f) \dot{x}^*(t_f))$$

$$= 2x^*(t_f) - \frac{1}{2} \dot{x}^{*2}(t_f)$$

$$x^*(1) = 4 = 1 + C_1 + C_2 \Rightarrow C_1 + C_2 = 3 \quad \text{... (1)}$$

$$x^*(t_f) = 4 = t_f^2 + C_1 t_f + C_2$$

$$2 \{ t_f^2 + C_1 t_f + C_2 \} - \frac{1}{2} \{ 2t_f + C_1 \}^2$$

$$2t_f^2 + 2C_1 t_f + 2C_2 - 2t_f^2 - 2C_1 t_f - \frac{1}{2} C_1^2 = 0 \quad \text{... (2)}$$

$$2C_2 - \frac{1}{2} C_1^2 = 0$$

$$\textcircled{1}, \textcircled{2}: 2(3 - C_1) - \frac{1}{2} C_1^2 = 0$$

$$-\frac{1}{2} C_1^2 - 2C_1 + 6 = 0$$

$$C_1^2 + 4C_1 - 12 = 0$$

$$(C_1 + 6)(C_1 - 2) = 0 \rightarrow C_1 = 2, -6$$

$$C_2 = 1, 9$$

$$\downarrow$$

$$\therefore x^*(t) = t^2 - 6t + 9$$

$$x^*(t_f) = t_f^2 - 6t_f + 9 = 4$$

$$(t_f - 1)(t_f - 5) = 0 \rightarrow t_f = 1, 5$$

$$t_f = 5$$

4) Ex 4-2-5

$$J(x) = \int_{t_0}^{t_f} [1 + \dot{x}^2(t)]^{1/2} dt$$

$$t_0 = 0 \quad x(0) = 0$$

$$t_f \text{ \& } x(t_f) \text{ free}$$

$$x(t_f) \text{ free on } \theta(t) = -5t + 15$$

Euler equation

$$0 = \frac{\partial g}{\partial x}(x^*(t), \dot{x}^*(t), t) - \frac{d}{dt} \left[\frac{\partial g}{\partial \dot{x}}(x^*(t), \dot{x}^*(t), t) \right]$$

$$= (1 + \dot{x}^2)^{1/2} - \frac{d}{dt} (\dot{x}^* (1 + \dot{x}^{*2})^{-1/2})$$

$$= \ddot{x}^* (1 + \dot{x}^{*2}) - \ddot{x}^* \dot{x}^{*2} = 0$$

$$\Rightarrow \ddot{x}^* = 0$$

$$x^*(t) = c_1 t + c_2$$

$$x^*(0) = 0 \rightarrow c_2 = 0$$

since $x(t_f)$ & t_f are related

$$\left[\frac{\partial g}{\partial x}(x^*(t_f), \dot{x}^*(t_f), t_f) \right] \left[\frac{d\theta}{dt}(t_f) - \dot{x}^*(t_f) \right] + g(x^*(t_f), \dot{x}^*(t_f), t_f) = 0$$

$$\left(\frac{\dot{x}^*}{(1 + \dot{x}^{*2})^{1/2}} \right) (-5 - \dot{x}^*) + (1 + \dot{x}^{*2})^{1/2} = 0$$

$$-5\dot{x}^* - \dot{x}^{*2} + 1 + \dot{x}^{*2} = 0$$

$$\therefore -5\dot{x}^* + 1 = 0 \rightarrow -5c_1 + 1 = 0 \rightarrow c_1 = \frac{1}{5}$$

$$x^*(t_f) = \theta(t_f)$$

$$\frac{1}{5} t_f = -5t_f + 15$$

$$\therefore t_f = \frac{15}{26} = 2.88$$

$$\therefore \text{sol: } x^*(t) = \frac{1}{5}t$$